

BUTTS AAF, CO, USAWMO: **724680**Lat: **38.678N**Lon: **104.757W**Elev: **5838**StdP: **11.85**Time Zone: **-7.00 (NAM)**Period: **07-19**WBAN: **94015****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	1.9	7.2	-9.0	4.2	25.7	-4.3	5.4	28.1	35.8	38.3	31.3	33.3	5.2	330	0.603

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	27.1	92.8	58.9	90.3	59.0	87.7	58.9	64.6	79.5	63.6	79.2	62.4	78.0	11.4	120

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
60.7	98.8	67.9	59.2	93.6	66.8	57.8	88.8	67.0	32.8	79.7	31.9	79.2	30.9	77.9	68.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
30.2	25.3	20.9	DB	-7.8	96.8	6.1	1.9	-12.2	98.1	-15.8	99.2	-19.2	100.3	-23.6	101.6
			WB	-8.3	67.5	6.0	1.0	-12.7	68.2	-16.2	68.8	-19.6	69.4	-23.9	70.2

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	51.0	31.8	33.1	42.3	47.9	56.2	68.6	72.7	70.5	64.4	51.2	41.1	30.8	(6)
	DBStd	17.19	10.16	11.15	9.80	8.86	8.28	6.17	4.58	4.49	6.53	9.59	9.85	11.02	(7)
	HDD50	2487	564	478	272	143	39	0	0	0	3	102	290	596	(8)
	HDD65	5770	1028	894	703	514	286	35	5	10	87	431	718	1059	(9)
	CDD50	2845	1	4	34	79	233	557	705	634	436	138	22	2	(10)
	CDD65	653	0	0	0	0	14	142	245	179	71	2	0	0	(11)
	CDH74	8457	0	1	14	39	366	2085	2747	1977	1095	131	2	0	(12)
	CDH80	3379	0	0	0	1	85	909	1262	771	340	11	0	0	(13)
	WSAvg	8.9	7.9	9.1	9.7	10.9	9.9	9.6	8.4	7.9	8.5	8.8	8.5	8.1	(14)
	PrecAvg	17.6	0.5	0.4	1.0	2.0	2.3	1.8	3.2	3.1	1.4	0.9	0.6	0.4	(15)
(16) Precipitation	PrecMax	25.7	1.2	1.5	2.5	7.2	8.3	5.1	6.3	5.5	4.8	2.6	2.4	1.4	(16)
	PrecMin	8.1	0.1	0.0	0.1	0.1	0.7	0.2	0.8	0.4	0.0	0.0	0.0	0.1	(17)
	PrecStd	4.3	0.3	0.3	0.6	1.5	1.7	1.3	1.5	1.3	1.1	0.7	0.5	0.3	(18)
(20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	65.2	69.0	76.1	78.6	86.7	96.8	95.1	93.5	91.1	81.9	73.3	64.6	(19)
		MCWB	43.5	44.5	47.3	49.1	53.4	57.6	59.5	58.9	59.2	51.5	46.6	42.7	(20)
	2%	DB	59.2	62.7	71.0	74.5	82.2	92.5	92.9	90.3	86.6	77.8	68.6	58.7	(21)
		MCWB	40.0	41.5	44.9	48.1	51.9	56.8	59.9	59.7	56.7	50.6	44.9	39.9	(22)
	5%	DB	54.5	58.3	66.3	70.9	78.2	89.2	90.6	87.6	84.1	74.1	64.0	54.2	(23)
		MCWB	38.0	39.7	43.0	46.6	51.8	56.4	60.1	60.1	56.4	49.7	43.5	37.9	(24)
	10%	DB	49.5	52.9	61.8	66.8	73.8	85.6	87.8	84.9	81.1	69.5	59.3	49.0	(25)
		MCWB	35.4	37.2	41.7	45.3	51.2	56.6	60.5	60.4	55.6	48.2	41.0	35.7	(26)
	0.4%	WB	43.4	45.2	48.0	52.7	58.3	65.0	66.3	66.2	62.7	56.2	47.7	43.7	(27)
		MCDB	62.4	67.0	72.8	67.5	74.7	80.5	79.0	80.4	79.6	69.7	69.1	62.7	(28)
(30) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	40.5	42.2	46.0	50.3	56.0	62.8	64.7	64.6	60.5	54.0	45.9	40.6	(29)
		MCDB	57.4	61.0	68.1	66.7	72.4	79.5	79.0	79.6	76.9	68.8	66.8	57.1	(30)
	5%	WB	38.3	40.0	44.0	48.5	54.4	61.0	63.8	63.5	59.2	51.4	43.9	38.2	(31)
		MCDB	53.7	57.5	64.1	65.7	70.4	77.8	79.4	78.4	76.1	68.2	62.7	53.3	(32)
	10%	WB	35.6	37.4	42.1	46.7	52.9	59.5	62.7	62.2	57.8	49.6	41.7	35.7	(33)
		MCDB	48.2	52.3	60.1	64.2	68.8	76.9	78.7	77.0	74.3	67.3	58.1	48.8	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	25.6	25.6	27.7	26.1	25.5	28.8	27.1	26.1	27.2	26.7	26.7	24.6	(35)
		MCDBR	32.4	33.6	34.2	33.1	33.2	34.2	31.9	30.7	31.8	33.9	33.4	32.8	(36)
		MCWBR	17.7	17.5	16.0	14.2	12.8	11.4	9.7	9.7	11.3	14.3	16.1	17.9	(37)
		MCDBR	31.1	33.1	32.5	30.0	28.2	28.1	25.9	25.2	27.6	29.0	31.5	31.3	(38)
	5% WB	MCWBR	17.4	17.6	15.6	13.5	11.9	11.5	9.3	9.6	10.6	13.3	15.6	17.7	(39)
(40) Clear-Sky Solar Irradiance	taub		0.235	0.239	0.261	0.287	0.306	0.320	0.350	0.336	0.300	0.266	0.244	0.237	(40)
	taud		2.624	2.642	2.576	2.512	2.518	2.495	2.466	2.495	2.569	2.640	2.626	2.613	(41)
	Ebn at Noon		311	322	321	315	308	302	292	295	302	306	304	302	(42)
	Edh at Noon		21	24	29	33	33	34	35	33	29	24	21	20	(43)
(44) All-Sky Solar Radiation	RadAvg		874	1176	1571	1897	2095	2328	2186	1994	1728	1305	957	770	(44)
	RadStd		45	75	115	115	156	107	92	71	89	76	54	41	(45)

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47) Regional (28 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page